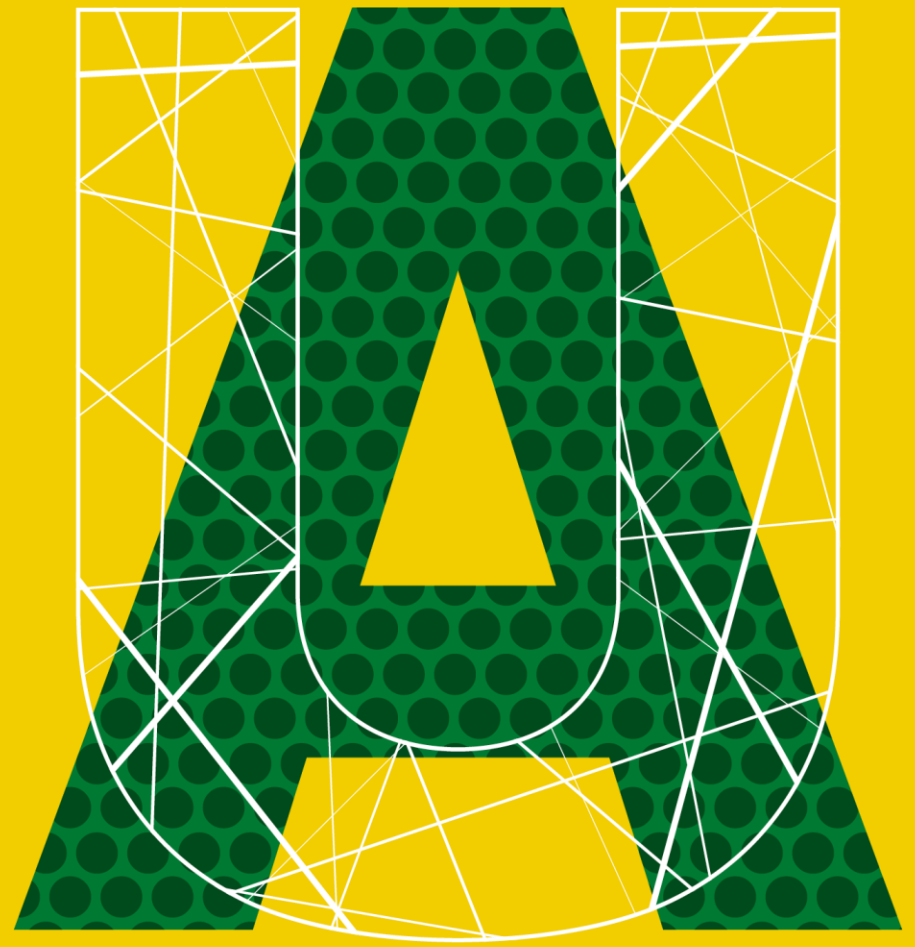
Effective Prediction Utilization:

Adaptive Execution and Self-Imitation-Based Fine-Tuning for Robotic Pushing

Qihang Feng

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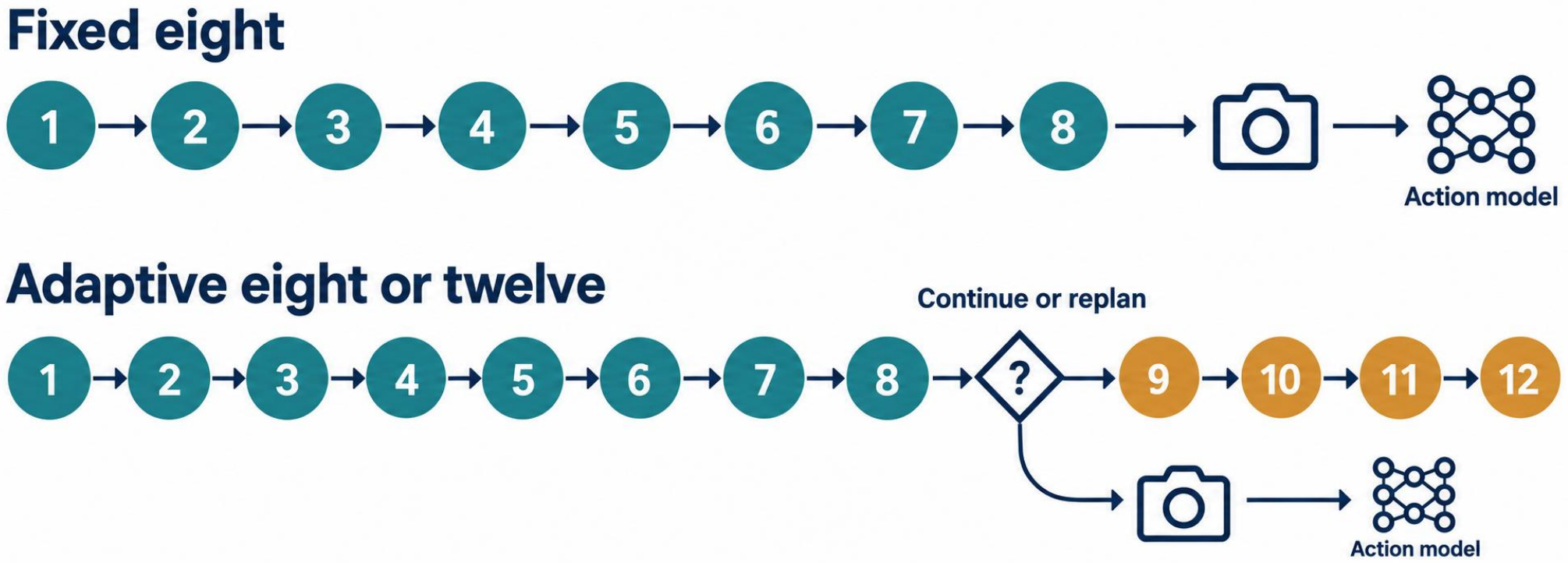


ECE 730 PROJECT PROPOSAL

Computation and AI for Cyber-Physical Systems

Prediction and execution

Each prediction requires computation. Remaining movements may still be useful.



The model generates the same sequence length in both cases

Signals for continuing or replanning

Signals: target coverage, decline, recovery and path smoothness

Continue up to twelve

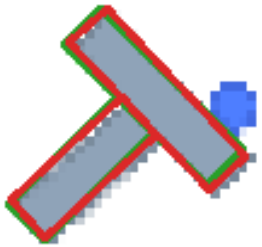
Clear visual estimate, limited recent decline,
and smooth remaining movements

Replan after eight

Near target, uncertain alignment,
or a failed continuation check

Interrupt early

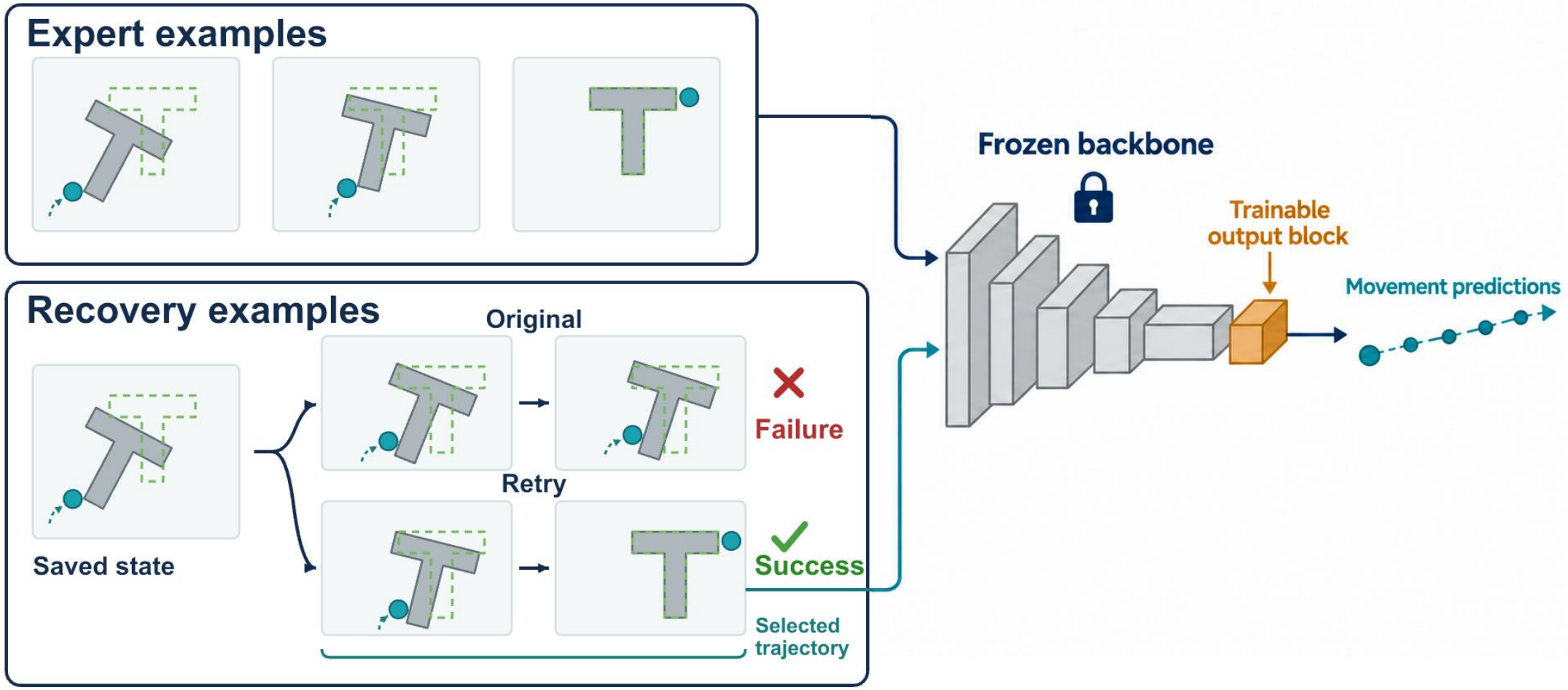
Repeated warnings, accumulated alignment loss,
and no clear recovery



A temporary alignment drop can be part of a useful adjustment

Self-imitation-based fine-tuning

Retry saved states from failed cases and retain successful continuations.



Update only the final convolution block. Keep other weights frozen.

Planned evaluation

Configuration	Model	Execution
Original baseline	Original pretrained	Fixed eight
Adaptive execution	Original pretrained	Adaptive 8/12
Fine-tuning	Fine-tuned	Fixed eight
Adaptive execution + fine-tuning	Fine-tuned	Adaptive 8/12

Matched scenes, repeated prediction samples, equal movement limits

Success (>95% coverage) and average final alignment
Average prediction calls and average complete-attempt runtime

Separate fine-tuning and evaluation scenes. Include failed attempts.

Expected contribution and scope

Target outcome

More successful pushes with fewer predictions and shorter runtime

Assess the effect on average final alignment

Report improvements, regressions and uncertainty

Scope: PushT simulation on a GTX 1650 laptop
Simulation pauses while the model predicts movements

Thank you!



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